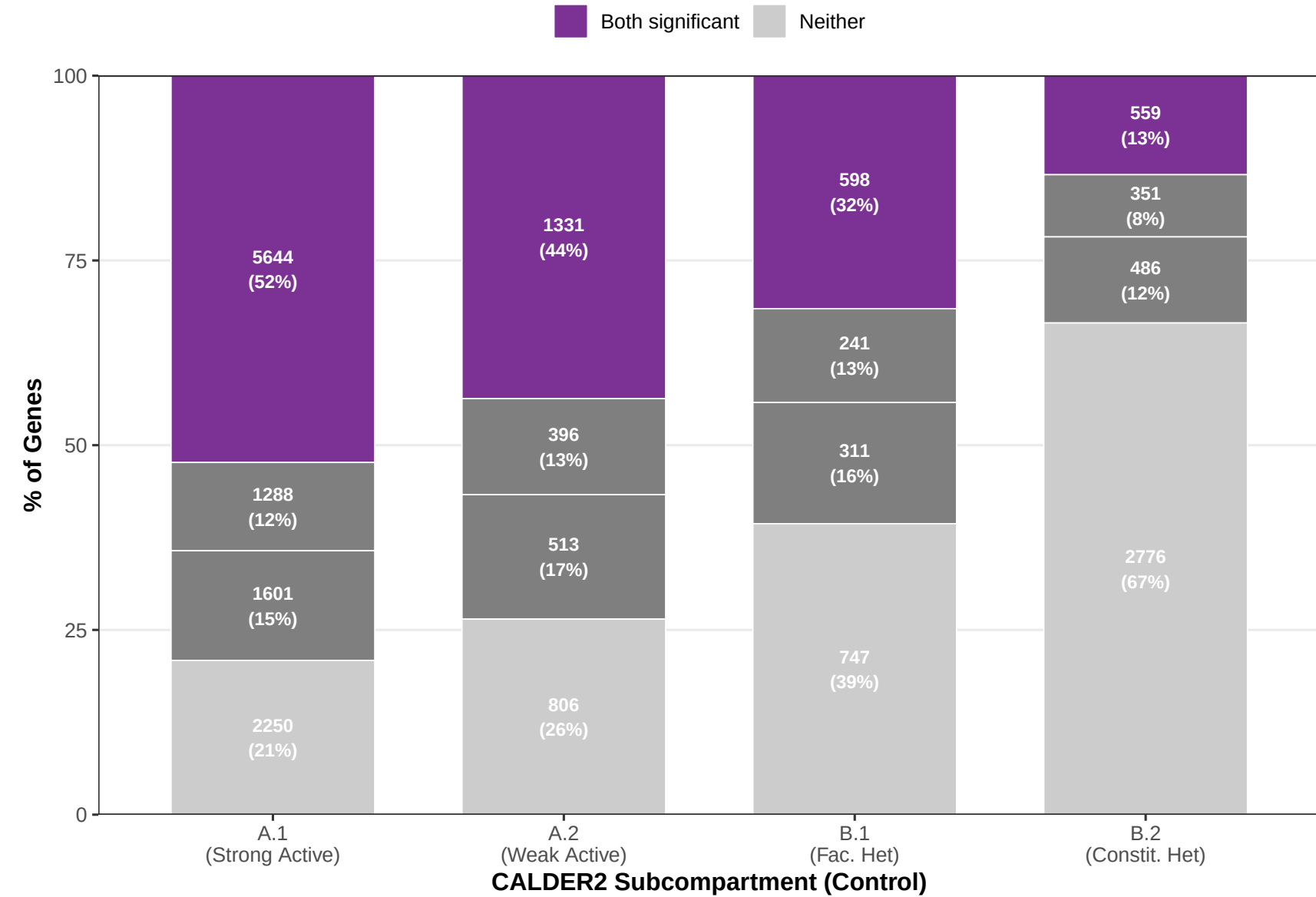


Compartment Shift Stoichiometry: 5hmC Is Redistributed, Not Removed

Near-stoichiometric mC/hmC balance + partial hmC retention + baseline-dependent depletion -> TET impediment

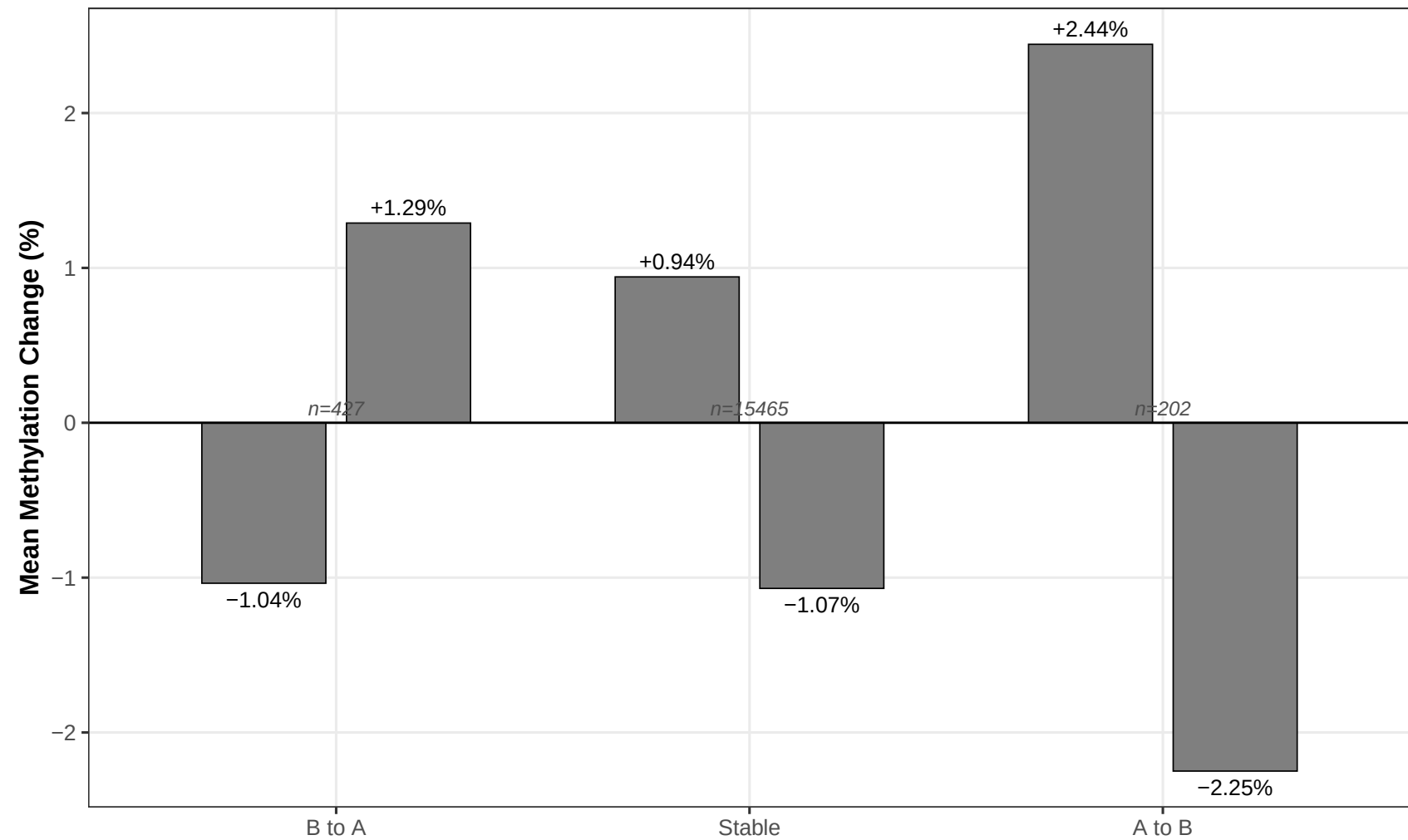
5mC / 5hmC Co-Significance by Subcompartment

Co-significant fraction drops monotonically from active (56%) to heterochromatic (19%)



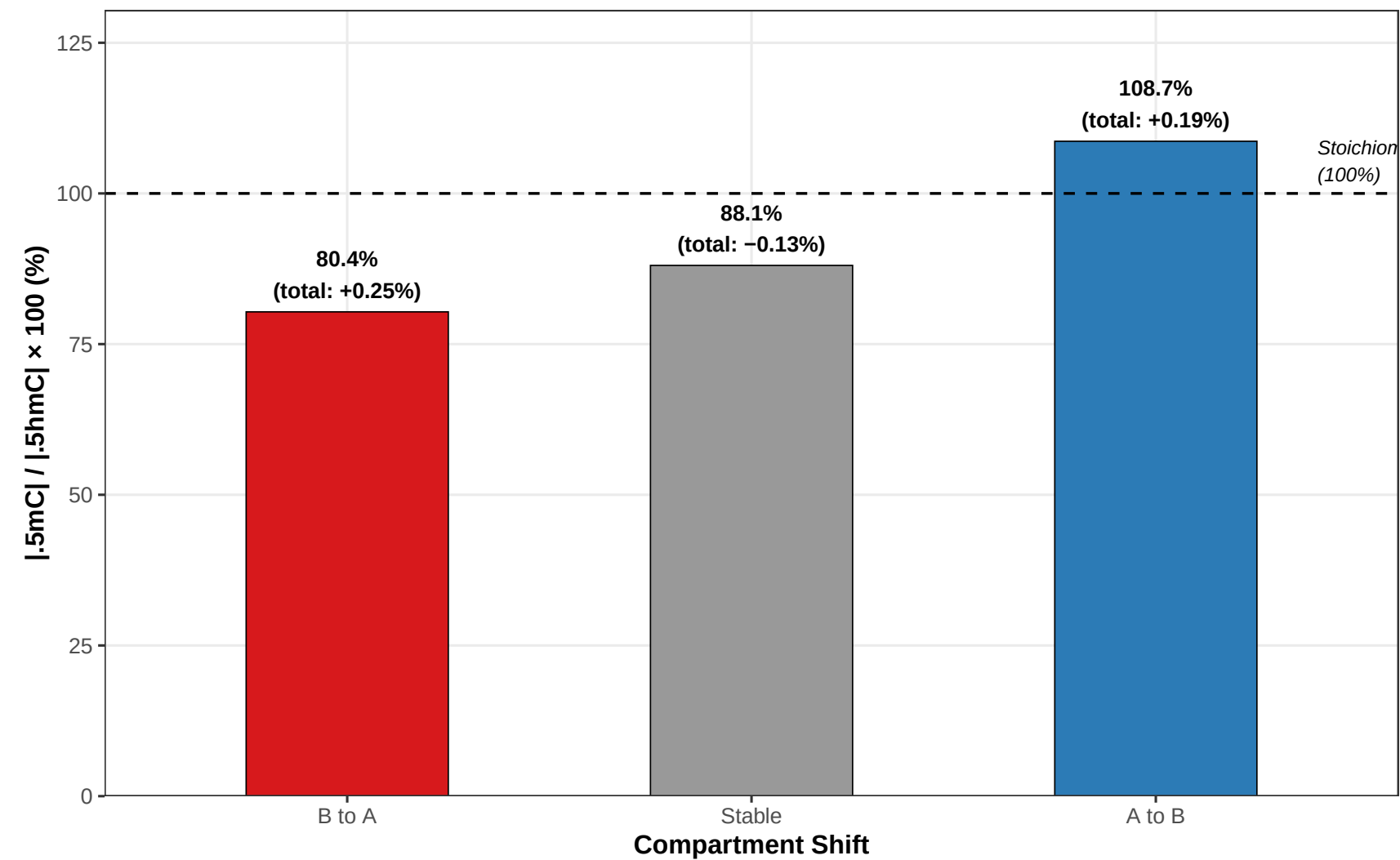
Methylation Change by Compartment Transition

A->B genes gain 2.4× more methylation than B->A genes lose | Mirror-image 5mC/5hmC at both



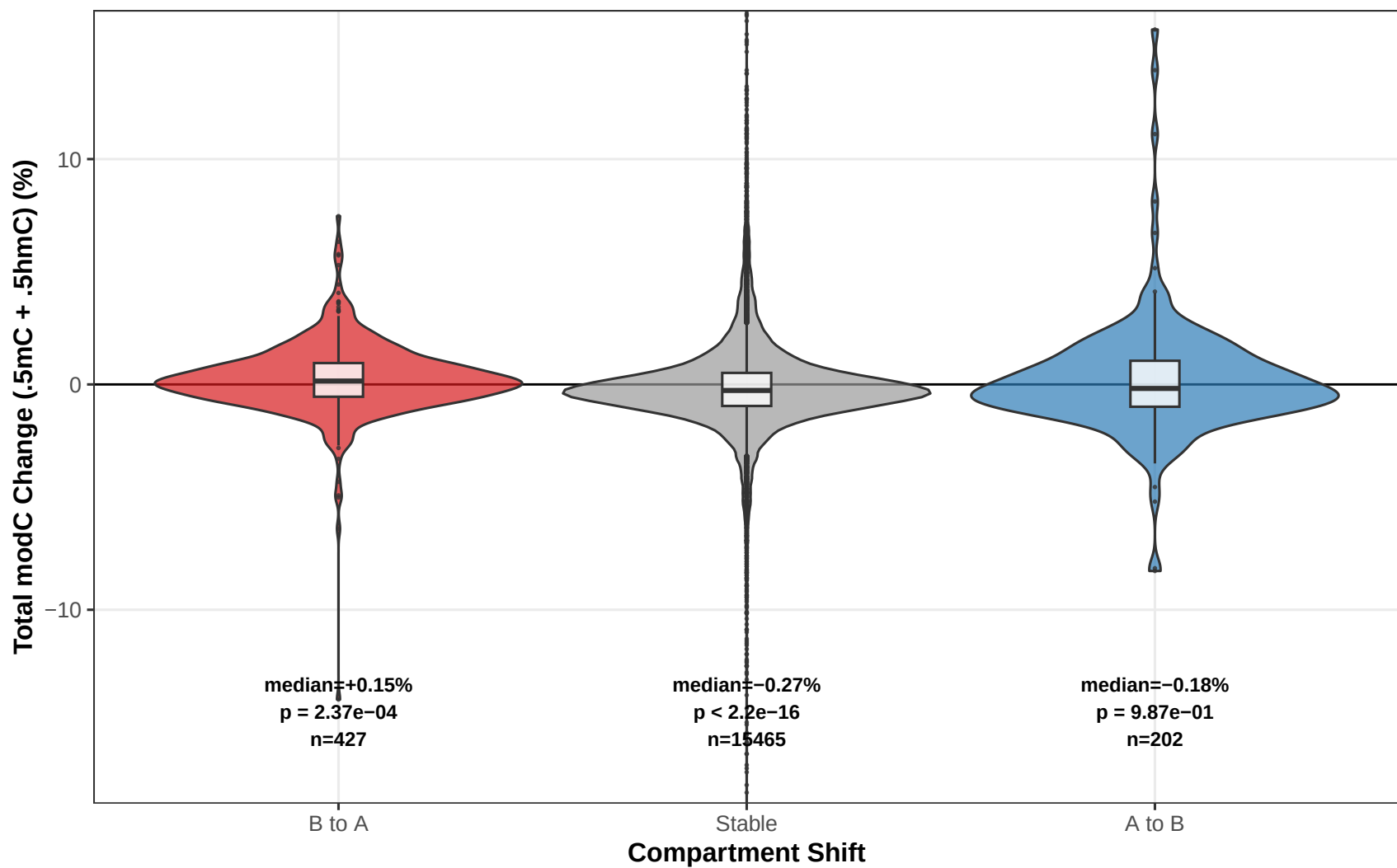
Stoichiometric Ratio: |.5mC| / |.5hmC|

A->B: 108.7% – 91% of 5hmC loss recaptured as 5mC (near-stoichiometric redistribution)



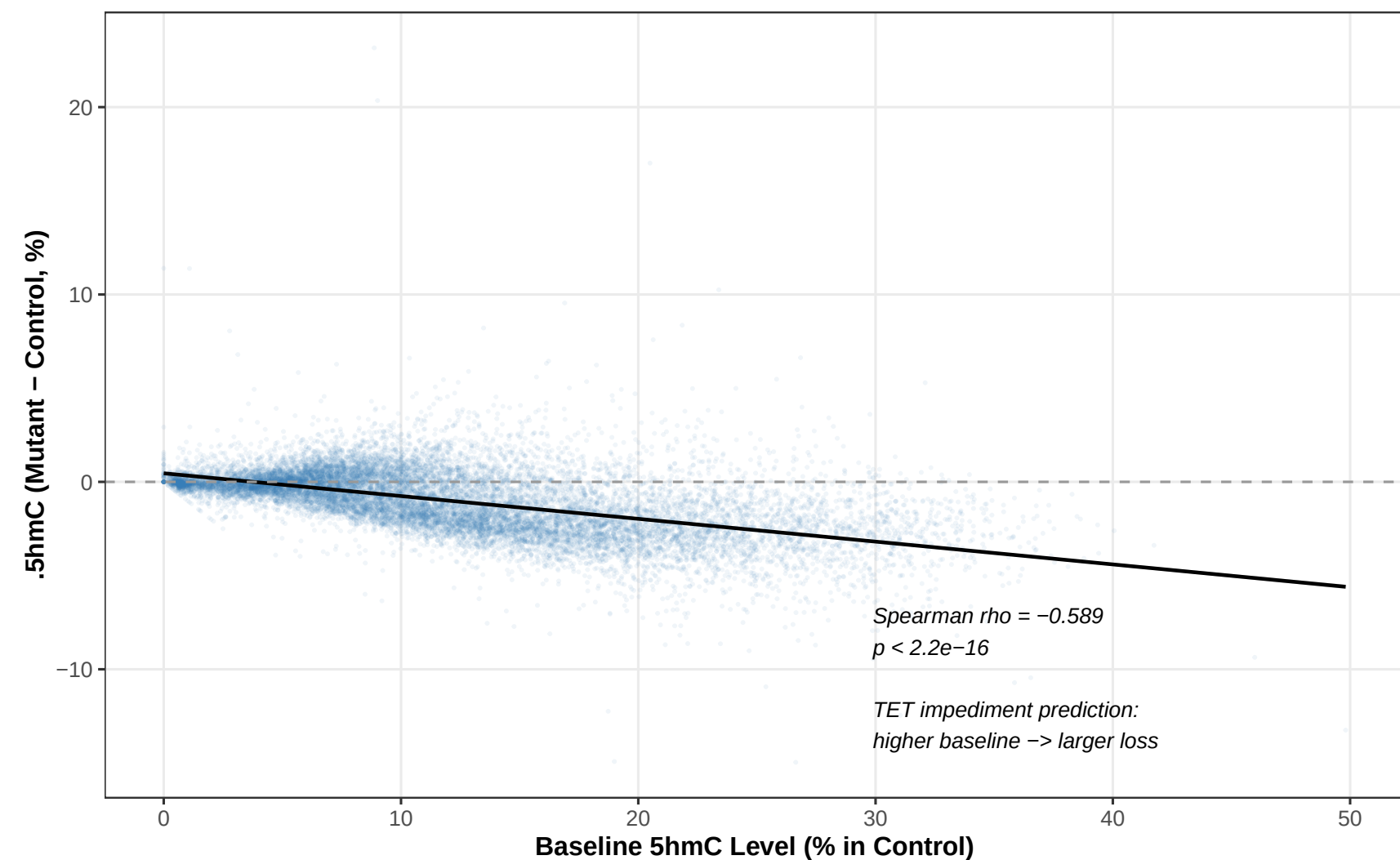
Total Modified Cytosine Change (.5mC + .5hmC) Per Gene

Centered near zero across all shifts – 5hmC loss is redistributed to 5mC, not removed



5hmC Change vs Baseline Level: TET Impediment Consistency

Genes with more 5hmC substrate show larger decreases – production blocked, not active removal



5hmC Levels: Control vs Mutant by Subcompartment

5hmC is reduced but NOT eliminated – consistent with blocked production, not active removal

